

Configuring Log Collection

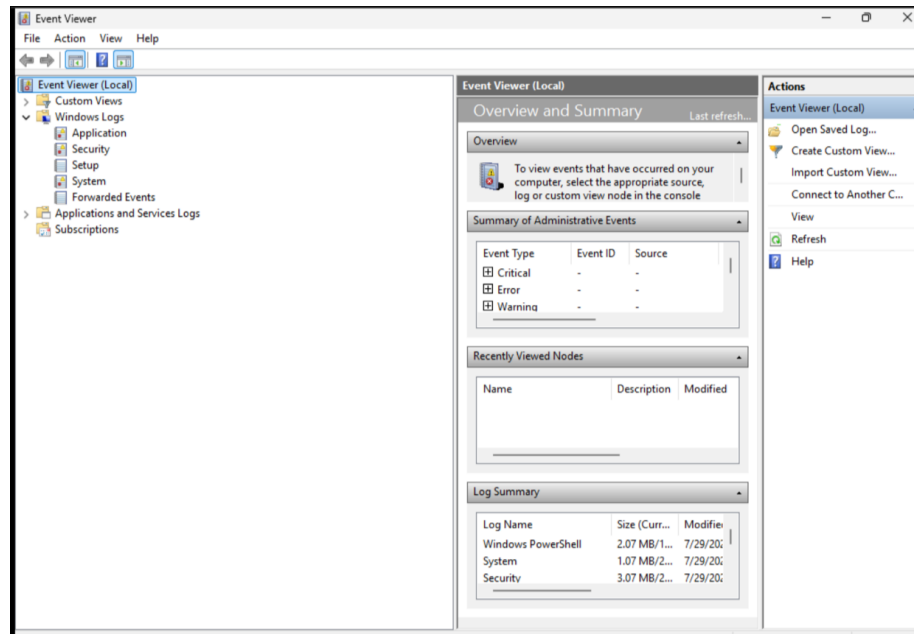
BELASRI Ayman

1 Introduction

Logs are timestamped records generated by systems and applications to track events, user actions, and errors acting as a digital audit trail. Centralizing these logs into a SIEM like Wazuh provides real-time visibility, allowing security teams to correlate events and detect threats across endpoints instantly.

2 Windows Event Log

On Windows endpoints, event logs are saved to the local disk in a binary format using `.evtx` files under `C:\Windows\System32\winevt\Logs\`. We can view these logs visually using the built-in **Event Viewer** (`eventvwr.msc`).



2.1 Categories of Windows Logs

- **Security:** Tracks logons, user privileges, and authentication events (e.g., Event ID 4625 for failed logins).
- **System:** Records OS operations, driver status, and hardware events (e.g., service crashes or reboots).
- **Application:** Stores events and error reports generated by third-party software.
- **Forwarded Events:** Stores events collected from remote endpoints if the machine acts as a central Windows Event Forwarding (WEF) server.
- **Setup :** Logs events related to Windows updates, patch deployments, and OS installations.

2.2 Key Components of a Windows Log

Each Windows log entry consists of structured metadata fields designed to give security analysts immediate context about system activities.

- **Event ID:** Unique code identifying the event type (e.g., 4624 for successful login).
- **Timestamp:** Exact date and time the event occurred.
- **Source:** Component or service that generated the log.
- **Severity:** Urgency ranking (*Info, Warning, Error, Critical, Audit Failure*).
- **User / Account:** Username or SID that executed the action.
- **Computer:** Hostname of the originating endpoint.
- **Event Data:** Action details like IP addresses, file paths, or process names.

2.3 Identifying Threat Activities via Event IDs

SIEM platforms map specific Event IDs to detect common adversary tactics across the network.

Activity	Source	Event ID	Key Detection
Failed Logon	Security	4625	Multiple failures indicate brute-force or password spraying attempts.
Successful Logon	Security	4624	Tracks user logins; watch for off-hours access or abnormal source IPs.

Activity	Source	Event ID	Key Detection
Process Creation	Sysmon	1	Detects execution of malicious commands, scripts, or abnormal parent-child processes.
Account/Privilege Change	Security	4720 / 4672	Flags new user account creation or assignment of administrative privileges.

3. Sysmon

System Monitor (Sysmon) is a free Windows system service from Microsoft Sysinternals that provides advanced telemetry on endpoint activity. Unlike standard Windows Event Logs, Sysmon focuses strictly on security-relevant behavior—capturing detailed context on process creation, network connections, file modifications, and registry changes to help SIEMs like Wazuh detect sneaky threats.

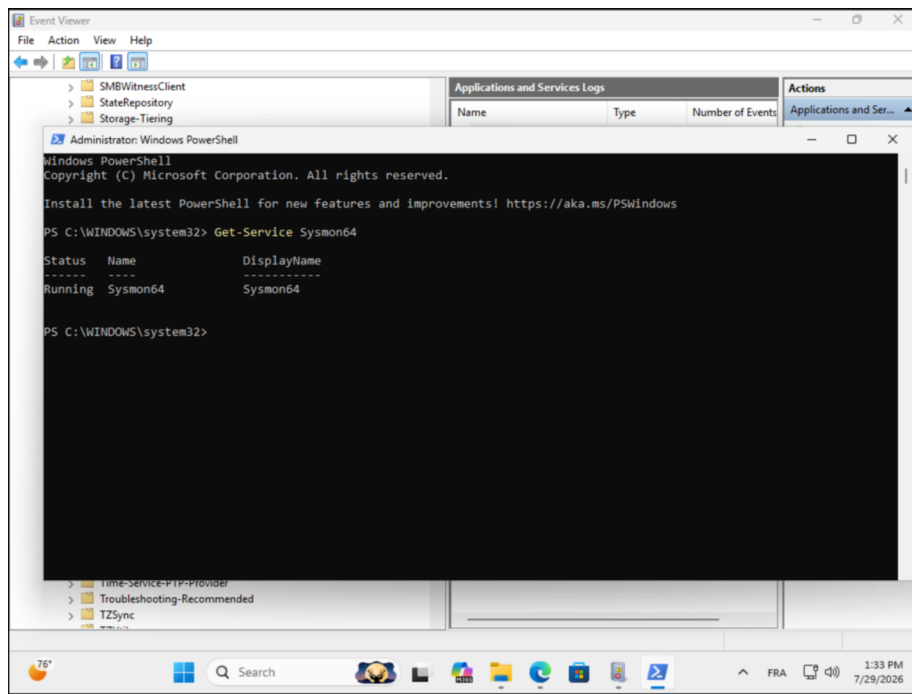
3.1 Sysmon Instalation

1. We download the latest Sysmon zip package from the official Microsoft Sysinternals site and extract it.
2. We then download a security-focused configuration file like **SwiftOnSecurity** (into the same folder.
3. Open PowerShell as Administrator, navigate to the folder, and run:

```
.\sysmon64.exe -accepteula -i sysmonconfig-export.xml
```

We can ensure the Sysmon service is running

```
Get-Service Sysmon64
```



3.2 Using Wazuh to monitor Sysmon events

Once Sysmon is installed on the endpoint, we need to configure the Wazuh agent to collect its logs and forward them to the Wazuh Manager. To do that we should :

1. Edit the local agent configuration file (`C:\Program Files (x86)\ossec-agent\ossec.conf`) on the Windows endpoint as Administrator and add the following block inside the `<ossec_config>` section:

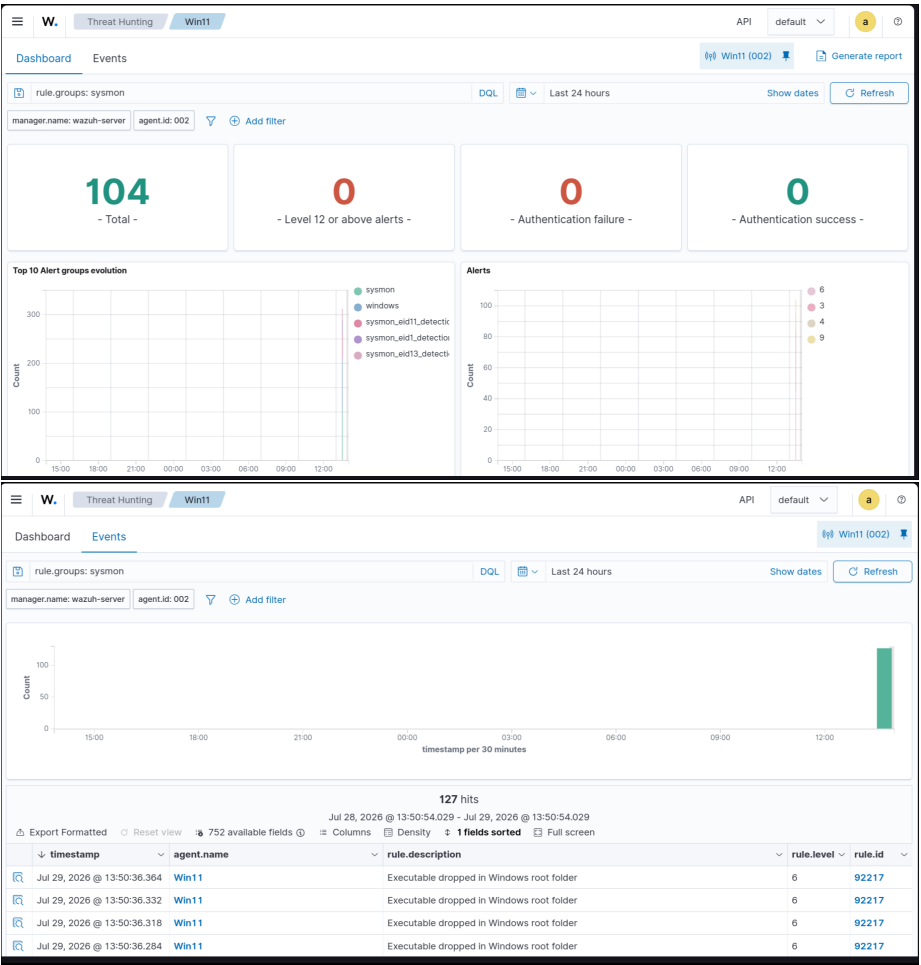
```
<localfile>
  <location>Microsoft-Windows-Sysmon/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
```

2. Restart the Wazuh Agent

Restart-Service Wazuh

3. To verify that its working properly , we can check the Wazuh Dashboard for sysmon events.To filter all sysmon events we use

```
text rule.groups: sysmon
```



Conclusion

By completing this setup, we turned basic Windows endpoint into active security sensor, giving us immediate, central visibility when anything fishy happens on the network.